

Database Management Completed SPARK Reference

IslandCare Medical Centre - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: IslandCare Medical Centre

Sector: Medicine

Questions being answered

Task A - Tables

1. Create a database named IslandCareDB_YourFullName.accdb.

How SPARK completed it

- Open your database software and create a blank database.
- Save it immediately with the exact filename in the question.
- Keep the database and all later SBA files inside one organised project folder.

Completed result

SPARK reference filename: IslandCareDB_YourFullName.accdb.

2. Create the Patient, Service and Appointment tables using suitable field names, data types and primary keys.

How SPARK completed it

- Create Patient first and enter the fields exactly as planned in your data dictionary.
- Repeat the process for Service and Appointment.
- Choose the primary key before entering records. Check currency, date and numeric data types carefully.

Completed result

Completed table structures:

Patient: PatientID - Short Text, primary key, FirstName - Short Text, LastName - Short Text, DOB - Date/Time, Gender - Short Text, Phone - Short Text, Parish - Short Text, EmergencyName - Short Text, EmergencyRelation - Short Text, EmergencyPhone - Short Text

Service: ServiceCode - Short Text, primary key, ServiceName - Short Text, StandardFee - Currency

Appointment: AppointmentID - Short Text, primary key, PatientID - Short Text, foreign key, ServiceCode - Short Text, foreign key, VisitDate - Date/Time, PaidAmount - Currency, Status - Short Text

3. Enter realistic records and apply the validation rules required by the project.

How SPARK completed it

- Enter enough varied records to make your queries meaningful.
- Use validation rules for fields with a limited set or range of allowed values.
- Test the validation by deliberately entering one invalid value and confirming that the software rejects it.

Completed result

The SPARK completed database uses fictional records with different categories, dates, fees and statuses so every query can be tested.

4. Create the required relationships and enforce the correct links between the tables.

How SPARK completed it

- Open the Relationships window.
- Add all tables and drag each primary key to the matching foreign key.
- Check the real-world relationship before selecting referential-integrity options.

Completed result

Patient.PatientID 1-to-many Appointment.PatientID; Service.ServiceCode 1-to-many Appointment.ServiceCode.

Task B - Queries and Form

1. Create a query called Daily Appointments. Show appointments for a date entered by the user. Include patient name, service and status.

How SPARK completed it

- Add only the table or tables required by the question.
- Add the requested fields to the query grid.
- Enter the criterion: VisitDate = [Enter visit date].
- Run the query and compare every returned record with the source data.

Completed result

Query name: Daily Appointments. Criterion used: VisitDate = [Enter visit date].

2. Create a query called Outstanding Accounts. Join Patient, Service and Appointment and calculate the balance.

How SPARK completed it

- Add the related tables and confirm the join lines are correct.
- Add the display fields requested in the question.
- Create the calculated field using: Balance: [StandardFee]-[PaidAmount].
- Run the query and calculate one record manually to verify the result.

Completed result

Query name: Outstanding Accounts. Calculation used: Balance: [StandardFee]-[PaidAmount].

3. Create Patient Information Form with Appointment subform and clear patient-contact headings.

How SPARK completed it

- Use the main record source for the parent form.
- Use the related table as the subform source.
- Add a clear title, useful instructions and navigation controls.
- Open several records and confirm the subform changes with the main record.

Completed result

Completed form design: Patient Information Form with Appointment subform and clear patient-contact headings.

Task C - Report

1. Create Service Income Report grouped by ServiceName with visit count, total fees and average fee.

How SPARK completed it

- Use a query as the report source when it already contains the correct fields or calculation.

- Apply the exact grouping and sorting required by the question.
- Add the required summary in the group or report footer.
- Preview the report and correct clipped headings, blank pages and unreadable columns.

Completed result

Completed report: Service Income Report grouped by ServiceName with visit count, total fees and average fee.

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Completed database structure

Patient

Field	Design
PatientID	Short Text, primary key
FirstName	Short Text
LastName	Short Text
DOB	Date/Time
Gender	Short Text
Phone	Short Text
Parish	Short Text
EmergencyName	Short Text
EmergencyRelation	Short Text
EmergencyPhone	Short Text

Service

Field	Design
ServiceCode	Short Text, primary key
ServiceName	Short Text
StandardFee	Currency

Appointment

Field	Design
AppointmentID	Short Text, primary key
PatientID	Short Text, foreign key
ServiceCode	Short Text, foreign key
VisitDate	Date/Time
PaidAmount	Currency
Status	Short Text

Relationships

Patient.PatientID 1-to-many Appointment.PatientID; Service.ServiceCode 1-to-many Appointment.ServiceCode.

Sample records

PatientID	FirstName	LastName	Age	ServiceCode	StandardFee	PaidAmount
P001	Marlon	Reid	67	GEN	6500	3000
P002	Keisha	Hall	41	WEL	8500	8500
P003	Noel	James	72	DIA	12000	5000
P004	Tiana	Morris	29	GEN	6500	6500
P005	Howard	Grant	65	WEL	8500	4000
P006	Safiya	Brown	36	DIA	12000	12000

Completed queries

Daily Appointments

Show appointments for a date entered by the user. Include patient name, service and status.

Criterion: VisitDate = [Enter visit date]

Outstanding Accounts

Join Patient, Service and Appointment and calculate the balance.

Calculation: Balance: [StandardFee]-[PaidAmount]

Completed form

Patient Information Form with Appointment subform and clear patient-contact headings.

Completed report

Service Income Report grouped by ServiceName with visit count, total fees and average fee.

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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